

HW #8 - Unit 0 Review

Date _____ Period _____

Sketch the graph of each line.

1) $7x + 3y = -9$

2) $y = -x + 3$

Write the slope-intercept form of the equation of each line given the slope and y-intercept.

3) Slope = $-\frac{5}{2}$, y-intercept = -3

Write the standard form of the equation of the line through the given point with the given slope.

4) through: $(0, -1)$, slope = undefined

Write the standard form of the equation of the line through the given points.

5) through: $(2, -3)$ and $(-2, 3)$

Write the slope-intercept form of the equation of the line described.

6) through: $(-2, -1)$, parallel to $y = x - 5$

Write the point-slope form of the equation of the line described.

7) through: $(5, 5)$, parallel to $y = \frac{6}{5}x - 3$

Evaluate each function.

8) $w(x) = x^2 - 2$; Find $w(-7)$

9) $h(x) = |x - 1| - 1$; Find $h(-6)$

Simplify.

10) $-4\sqrt{20}$

11) $-6\sqrt{50}$

$$12) -6\sqrt{27x^4}$$

$$13) 7\sqrt{16x^4y^3}$$

Simplify. Your answer should contain only positive exponents.

$$14) \frac{-y^{-4} \cdot (y^{-3})^{-4}}{-x \cdot -x^{-3}y^3}$$

$$15) -\frac{y^4 \cdot 2xy^{-1}}{(-yx^2)^3}$$

Evaluate each expression.

$$16) (((-1) - 4 - 3) \cdot 2) \div (-4)$$

$$17) 2 + (-6) - (-6) - \frac{16}{(-2) \cdot (-2)}$$

Solve each equation.

$$18) -40 + 7n = 6(1 + 5n)$$

$$19) 2 - 6a = -4(7 - a)$$

Simplify each expression.

$$20) \frac{5}{2x} - \frac{5x}{4}$$

$$21) \frac{6m}{5mn} + \frac{3n}{5n}$$

Graph each equation.

$$22) y = 3|x| - 1$$

$$23) y = 2|x + 4| - 4$$

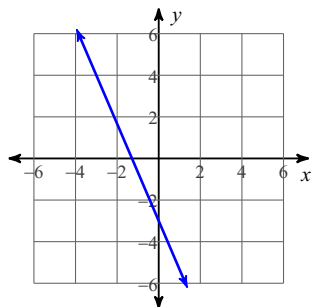
Sketch the graph of each function.

$$24) f(x) = \begin{cases} -2x + 2, & x < 4 \\ x - 2, & x \geq 4 \end{cases}$$

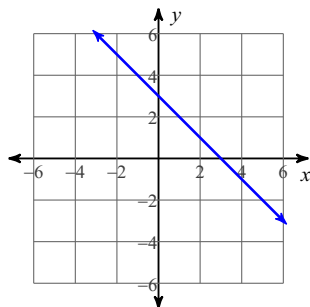
$$25) f(x) = \begin{cases} x + 4, & x \leq -4 \\ -|x|, & -4 < x \leq 1 \\ -5, & x > 1 \end{cases}$$

Answers to HW #8 - Unit 0 Review

1)



2)



$$3) y = -\frac{5}{2}x - 3$$

4) $x = 0$

5) $3x + 2y = 0$

6) $y = x + 1$

7) $y - 5 = \frac{6}{5}(x - 5)$

8) 47

9) 6

10) $-8\sqrt{5}$

11) $-30\sqrt{2}$

12) $-18x^2\sqrt{3}$

13) $28x^2y\sqrt{y}$

14) $-x^2y^5$

15) $\frac{2}{x^5}$

16) 4

17) -2

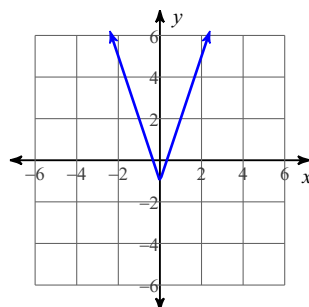
18) $\{-2\}$

19) $\{3\}$

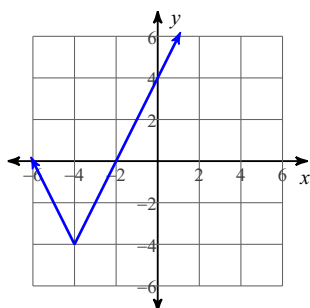
20) $\frac{10 - 5x^2}{4x}$

21) $\frac{6 + 3n}{5n}$

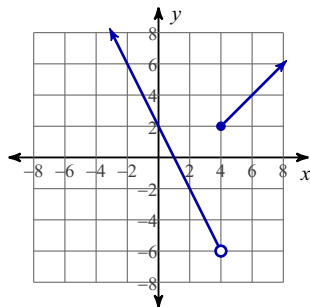
22)



23)



24)



25)

